

# KARTIKEYA GAUR

Centre for Infrastructure, Sustainable Transportation and Urban Planning (CiSTUP)

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[LinkedIn Profile](#)

## EDUCATION

|                                                                                   |                           |
|-----------------------------------------------------------------------------------|---------------------------|
| <b>Indian Institute of Science (IISc)</b>                                         | Bengaluru, India          |
| <i>M.Tech. in Smart Mobility and Logistics Systems (Current CGPA: 7.90/10.00)</i> | 2025 – Present            |
| <b>National Institute of Technology (NIT) Hamirpur</b>                            | Himachal Pradesh, India   |
| <i>B.Tech. in Civil Engineering (CGPA: 8.48/10.00)</i>                            | 2020 – 2024               |
| <b>St. Paul's Senior Secondary School</b>                                         | Jodhpur, Rajasthan, India |
| <i>Higher Secondary Certificate (Class XII - CBSE, 95.4%)</i>                     | 2019 – 2020               |
| <b>St. Paul's Senior Secondary School</b>                                         | Jodhpur, Rajasthan, India |
| <i>Secondary School Certificate (Class X - CBSE, 91.6%)</i>                       | 2017 – 2018               |

## RESEARCH & PROFESSIONAL EXPERIENCE

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Indian Institute of Remote Sensing (IIRS), ISRO</b>                                                                                                                                                                                                                                                                                                                                                                                             | Dehradun, India       |
| <i>Junior Research Fellow</i>                                                                                                                                                                                                                                                                                                                                                                                                                      | Sept 2024 – July 2025 |
| <ul style="list-style-type: none"><li>Developed an automated framework for river discharge and bathymetry estimation by integrating multi-mission satellite altimetry (ICESat-2, SWOT, Sentinel-3) using DBSCAN clustering and Piecewise NLSB algorithms.</li><li>Conducted extensive ADCP-based field validation across 12 Ganga River sites, validating rating curves and achieving a bathymetry prediction accuracy (RMSE) of 2.40 m.</li></ul> |                       |
| <b>BISAG-N</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | Gandhinagar, India    |
| <i>Geospatial Project Trainee</i>                                                                                                                                                                                                                                                                                                                                                                                                                  | Dec 2022              |
| <ul style="list-style-type: none"><li>Proposed three railway alignments in the mountainous terrain of Northeast India under the PM GatiShakti National Master Plan.</li><li>Analyzed terrain characteristics and topology in QGIS to generate shapefiles and model discharge points to support check-dam planning.</li></ul>                                                                                                                       |                       |
| <b>Defence Laboratory (DRDO)</b>                                                                                                                                                                                                                                                                                                                                                                                                                   | Jodhpur, India        |
| <i>Technical Trainee</i>                                                                                                                                                                                                                                                                                                                                                                                                                           | Jun – Jul 2022        |
| <ul style="list-style-type: none"><li>Evaluated military trafficability in arid regions of Jaisalmer under the supervision of Dr. S.L. Borana.</li><li>Combined supervised LULC classifications, DEM derivatives, and digitized soil maps in ArcGIS to generate a weighted overlay trafficability map.</li></ul>                                                                                                                                   |                       |

## SELECTED PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                               |                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Scalable Algorithm for Public Transit Skims</b>                                                                                                                                                                                                                                                                                                                                                            | <i>M.Tech Thesis</i>                       |
| <i>Supervisors: Prof. Tarun Rambha &amp; Prof. Abdul R. Pinjari</i>                                                                                                                                                                                                                                                                                                                                           |                                            |
| <ul style="list-style-type: none"><li>Developing a scalable C++ routing framework designed to generate transit network skim matrices using multi-criteria Pareto-optimal path algorithms, aiming to simulate millions of agent journeys efficiently for Bengaluru.</li></ul>                                                                                                                                  |                                            |
| <b>Intra-Urban Heat Island Effect Analysis: Case Study of Chandigarh</b>                                                                                                                                                                                                                                                                                                                                      | <i>BTech Project</i>                       |
| <ul style="list-style-type: none"><li>Leveraged USGS Landsat-8 thermal imagery (2003–2023) in Google Earth Engine to analyze Land Surface Temperature (LST) variations and quantify the Intra-Urban Heat Island effect, observing a long-term temperature rise of 2.80°C in Chandigarh (<b>0.14°C/year</b>) and a higher surge of 6.77°C in Jalandhar (<b>0.34°C/year</b>).</li></ul>                         |                                            |
| <b>Last-Mile Connectivity Analysis and Visualization</b>                                                                                                                                                                                                                                                                                                                                                      | <i>Multi-Modal Mobility Systems</i>        |
| <ul style="list-style-type: none"><li>Analyzed large-scale commuter survey data across Bengaluru, Delhi, and Nagpur last-mile connectivity patterns using python libraries and developed interactive Power BI dashboards.</li><li>Mapped Delhi's <b>high non-walk dependency zones</b> (13.35% area, 38/290 wards) and identified Nagpur has maximum population accessibility per station at 0.88%.</li></ul> |                                            |
| <b>Capstone Project: Warehouse Optimization at Land Port Petrapole</b>                                                                                                                                                                                                                                                                                                                                        | <i>Partner: Ernst &amp; Young (EY) LLP</i> |
| <ul style="list-style-type: none"><li>Formulated physical and process interventions to address a 191.7% warehouse over-utilization rate.</li><li>Modeled a Selective Pallet Racking system using ABC-XYZ inventory classification and proposed a 5-layer IoT data architecture.</li></ul>                                                                                                                     |                                            |

- Constructed a discrete-event simulation (DES) using AnyLogic to analyze vessel/vehicle queuing bottlenecks.

### Implementation of a Convex Travel Demand Forecasting Model

*Choice Modelling*

- Reproduced a combined travel demand optimization model (trip distribution, modal split, and traffic assignment) as a single convex program using Julia/JuMP.
- Reformulated conditional entropy terms into exponential cones and link-latency integrals into power cones for efficient resolution using the MOSEK solver.

### Nighttime Street Light Audit Using Computer Vision

*Edge AI*

- Deployed an edge-ready computer vision pipeline using YOLOv5 and YOLOv8 models trained on a custom 2.1k frame dataset.
- Optimized inference speeds on resource-constrained hardware using post-training quantization (FP16 and INT8), pruning, and knowledge distillation.

### Contributor Verification using Robust Deep Neural Networks

*ML for Cyber-Physical Systems*

- Developed a binary classification model on a dataset of 88,340 samples using the Keras Functional API.
- Implemented a multi-branch Deep Neural Network (DNN) with L2 regularization and Dropout to minimize generalization error, achieving a final Private Leaderboard Rank of 11/25 (Private Score: 0.8607).

## PUBLICATIONS

- Gaur, K., et al. (2025). “**SWOT Mission with Wide-Swath Altimetry: Observations and Insights into India’s Inland Waterbodies.**” *Journal of the Indian Society of Remote Sensing*.

## TECHNICAL SKILLS

- **Programming & Data Analysis:** Python (NumPy, Pandas, Scipy), C++, R, Julia/JuMP.
- **Machine Learning & Optimization:** TensorFlow, PyTorch, Keras, Scikit-learn, MOSEK.
- **Geospatial & Design Tools:** ArcGIS, QGIS, GEE (Google Earth Engine), ENVI, AutoCAD.
- **Simulation & Visualization:** SimPy, AnyLogic, Pygame, Power BI.
- **Languages:** English (fluent), Hindi (native), French (A1).

## KEY COURSES

- Machine Learning for Cyber-Physical and Mobility Systems; Transportation Demand and Supply Modelling; Introduction to Multi-Modal Mobility Systems; GIS and Remote Sensing; Network Science and Optimisation; Choice Modelling; Edge AI; Introduction to Algorithms and Software Programming.

## ACHIEVEMENTS & HONORS

- **Secured All India Rank (AIR) 10** in GATE 2023 (Geomatics Engineering) with an 819 score.
- **Secured 98.5 Percentile** in JEE 2020 (Engineering Entrance Examination).

## CO-CURRICULAR & LEADERSHIP ROLES

- **Student Auditor** — *Literacy Mission, NIT Hamirpur* **May 2022 – May 2024**  
Managed financial transparency and records for a student-run organization tutoring and supporting over 80 underprivileged children of migrated laborers.
- **Coordinator** — *C-Helix, Nimbus Technical Festival, NIT Hamirpur* **Dec 2021 – Jul 2023**  
Built and demonstrated an origami-based emergency shelter prototype intended for nomadic and road-side dwellers. Organized a hands-on QGIS training workshop for over 100 undergraduate participants.

## WORKSHOPS ATTENDED

- COEAI-SPARC Workshop on Hybrid Physics-AI Models for Climate, Weather and Water, IIT Kharagpur, June 2024.
- High-End Workshop (Karyashala) on “Mathematical Foundations for AI/ML,” SERB, IIT Indore, MP, May 2024.
- Remote Sensing and GIS in Rural Development, NPTEL, January 2023.

Certificates can be found [here](#).